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Unit 2 : Session 4 : SLO 2 | Array Implementation of Lists

Programming Practice II

I. Sum and Average

- `int sum = 0; for (i = 0; i < n; i++) sum += a[i];`
- `float avg = (float) sum / n;`

II. Armstrong Numbers

- `for (i = 0; i < n; i++) { int x = a[i], s = 0, t = x;`
- `while (t) { int d = t % 10; s += d * d * d; t /= 10; }`
- `if (s == x) printf("%d is Armstrong\n", x); }` // e.g. 153, 370, 371, 407

III. Decimal to Octal

- `int oct[32], k = 0, x = num;`
- `while (x) { oct[k++] = x % 8; x /= 8; }`
- `while (k) printf("%d", oct[--k]);` // prints octal digits in reverse

IV. Sum of Even Numbers

- `int s = 0; for (i = 0; i < n; i++) if (a[i] % 2 == 0) s += a[i];`

V. Search an Element (Linear Search)

- `int found = -1; for (i = 0; i < n; i++) if (a[i] == key) { found = i; break; }`
- `if (found >= 0) printf("Found at index %d", found); else printf("Not found");`